Hello, I'm Suvanshi Lakhotiya, jack of all trades, master of some. My hardware experience is right now limited to making the XRL8 bot (and being in top 5 out of almost 250 teams) which is what piqued my interest in hardware. I specifically had done the wiring and debugging in the whole process.

So, from what I understand about HackerFab, it is a silicon chip fabrication lab, working to make silicon chips at an economic price. I'm gonna be honest and say that I did not until very recently know what HackerFab was or how silicon chips are made or any of it; but I did some research, watched a few youtube videos, talked a bit to Claude and it does seem like a pretty interesting thing. I mean, it's astounding the scale at which these chips are made (both how small they are and the literal number of them required).

In the short term, I would like to explore what we are currently doing at HackerFab, what are the parts we have gotten working well, where are we hitting a snag in the ongoing projects and work on the lithography problem as I have explained below. Also, photoresist is baked on a hot plate before and after exposure, which is why I have chosen that as my topic out of the three options.

I would be interested in building the lithography tool used to expose the photoresist on the wafer to make a pattern. In order to expose the patterns using UV light, we can use a DMD (digital micromirror device) projector. It has millions of tiny mirrors which can be tilted in specific patterns in order to make new designs for each layer. Normally DMD projectors work only for visible light but there are special UV light DMD projectors which we can use. The slight hitch with this method is that it covers only a small area at a time, so the wafer needs to be moved very carefully between each design. To achieve this, we can use a sliding platform (which is accurate within 0.1 μm as we need it to move a few millimetres between tiles but it must land accurate to 0.1 μm, so the tiles join up), made of a low-expansion metal and a heavy base (to prevent vibrations) to move the wafer between each exposure. For higher accuracy, we can also use piezo actuators which give us much more precision but also happen to be more expensive. Both of these tools (the DMD projector and the sliding plate) are relatively affordable compared to actual lithography machines.

In fact, we can keep multiple DMD projectors side by side so that a whole row of patterns can be exposed at once (more heads add alignment errors between tiles, but save time). I found that the HackerFab group at CMU has already built a similar lithography setup (for about $3000). I think we could replicate a similar setup (after thoroughly going through their design and documentation of course) here for an even lower cost by using parts sourced in India (like the frame and the stage).

I haven't figured out how to tune the multiple DMD projectors to align and calibrate so their tiles join up perfectly or how we would hold the wafer at the exact focus height but I'm looking forward to brainstorming these problems.

In the long term, I would like to make my first transistor (from scratch) here.

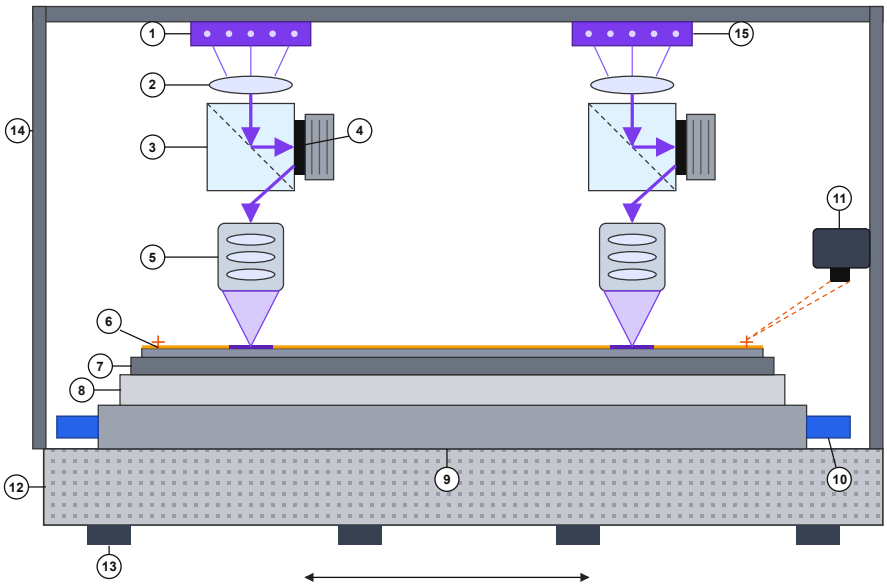
Going forward I would like to become an integral part of the team, and help other students explore this new domain. I would like to learn more about how we make these tiny transistors and put my problem-solving skills to the test by thinking of out-of-the-box ideas to make fabrication both cheaper and easier.

Shown below is a basic sketch of my idea

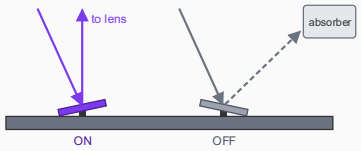
Maskless UV lithography stepper (DMD-based): concept design

Schematic, not to scale. The light path is simplified (a real TIR prism has different geometry).

Side view

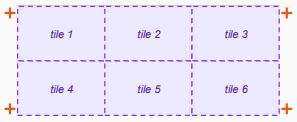


DMD close-up (each mirror = 1 pixel)



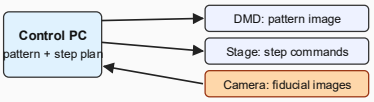
ON: light goes to the lens, so that spot is exposed
OFF: light goes to the absorber, so that spot stays dark

Step and repeat (stitching tiles)



Each tile is one DMD exposure. The stage moves between tiles.
Orange + = fiducial marks the camera uses to line things up.

Control



Callout key

1. UV LED light source (near-UV)
2. Illumination optics (spreads light evenly)
3. TIR prism (guides light to and from the DMD)
4. UV-capable DMD chip (millions of micromirrors)
5. Projection lens (shrinks the image)
6. Wafer or chip with photoresist layer
7. Vacuum chuck (holds the wafer flat)
8. Z stage (sets focus height)
9. XY stage (low-expansion material)
10. Fine actuators (stepper or piezo) for steps
11. Alignment camera (reads fiducial marks)
12. Heavy granite base (damps vibration)
13. Vibration-isolation feet
14. Rigid frame holding the projector heads
15. Second projector head (modular array)